



National University of Computer and Emerging Sciences, Lahore Campus

Course: Signals & Systems
 Program: BS(Electrical Engineering)
 Duration: 60 Minutes
 Paper Date: 8th April, 2017
 Section: ALL
 Exam: Midterm-II

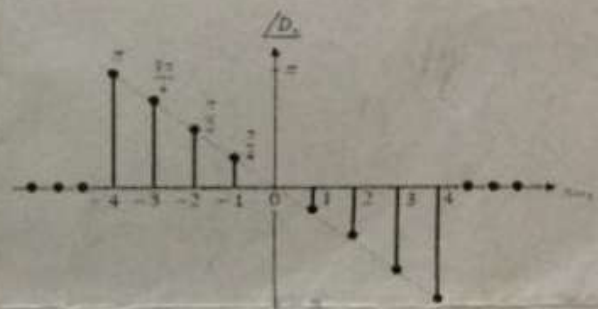
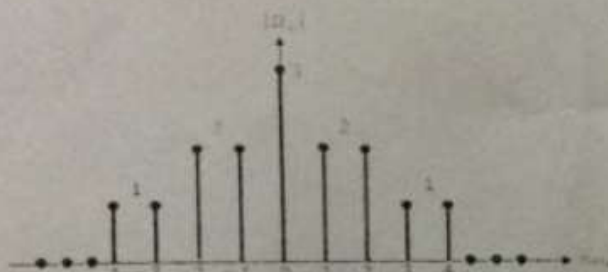
Course Code: EE223
 Semester: Spring 2017
 Total Marks: 40
 Weight: 15%
 Page(s): 1
 Roll No: [REDACTED]

Instruction/Notes: Carefully read the paper and attempt each question (only once). Attempt all parts of a question together.

Question #1: CLO (3)

15 Marks

Exponential Fourier Spectra of a periodic signal $x(t)$ are given below:



- Find the exponential Fourier series using the above spectra.
- Find the compact trigonometric Fourier series of $x(t)$.
- Sketch the spectra of compact trigonometric Fourier series using part (b).
- Find the bandwidth of the signal.

10 Marks

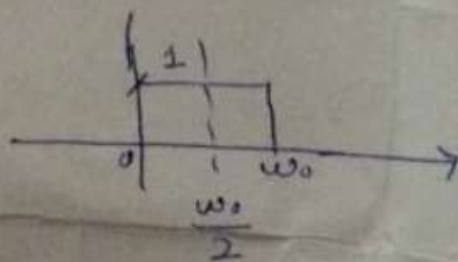
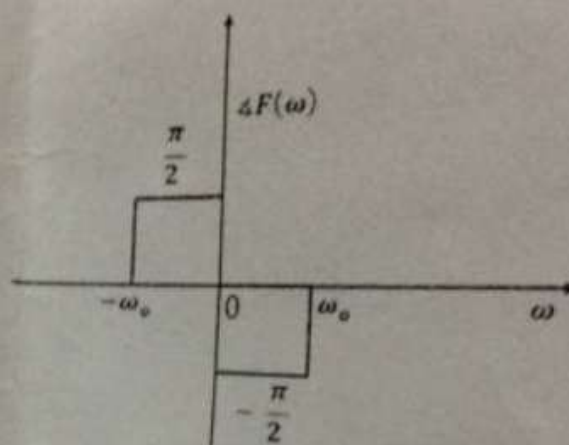
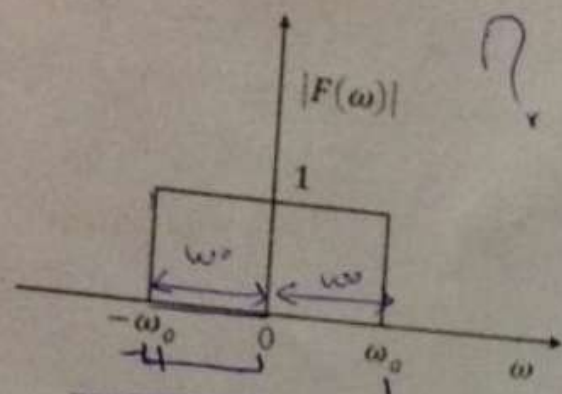
Question #2 CLO (3)

For the following continuous time periodic signals:

- $x(t) = \sin 50\pi t + \cos 30t$
- $y(t) = 3 + 2\cos\left(\frac{5\pi}{2}t\right) + \sin\left(\frac{6\pi}{5}t + \frac{\pi}{3}\right) + 7\cos\left(\frac{\pi}{4}t + \frac{\pi}{4}\right)$ $T=8?$
 - Find the fundamental frequency and time period. $1, T=2\pi?$
 - Identify which harmonics are present in the signal. $\pi/4, T=8?$

Question #3 CLO (4)

Find and simplify the Inverse Fourier transform $f(t)$ from the magnitude and phase spectra given below



$$\left[\frac{\omega - \frac{\omega_0}{2}}{2} \right]$$